

18.06 — Lecture 1

n Linear Equations, n Unknowns: Row Picture, Column Picture, Matrix Form

Course: MIT 18.06 Linear Algebra
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Topics

1. n linear equations, n unknowns
2. Row picture
3. Column picture
- 4,5. Matrix form

1 2 Equations, 2 Unknowns

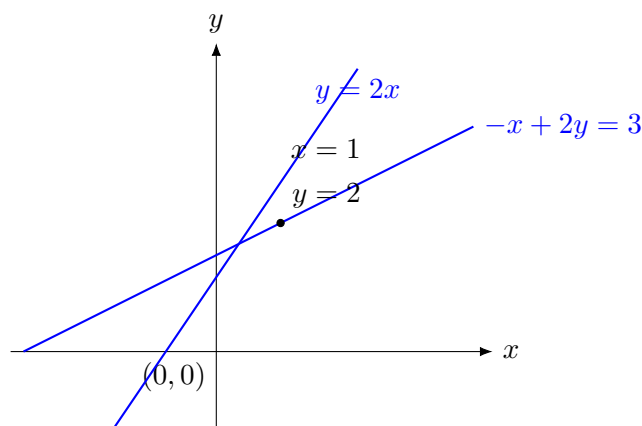
$$\begin{aligned}2x - y &= 0 \\ -x + 2y &= 3\end{aligned}$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$AX = b \quad \text{— vectors/matrices}$$

1.1 Row Picture

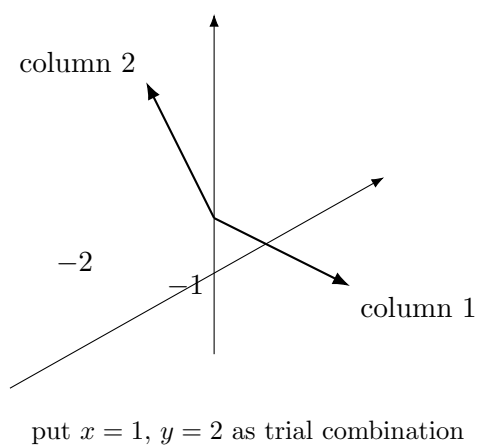
Each row of the system, taken as an equation, describes a *line* in the x - y plane. The row picture is the picture of these lines and their intersection.



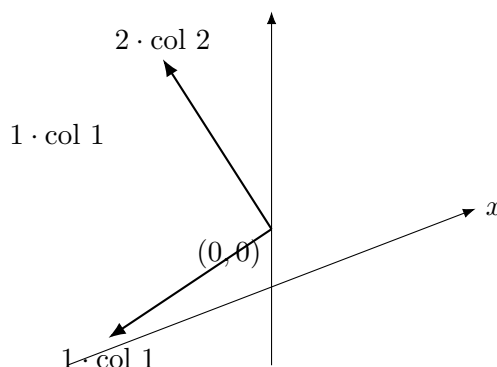
— solution to the 2 equations

1.2 Column Picture

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix} \quad \text{— linear combination of columns}$$



2 Linear Combinations Fill the Plane



representing each column as a separate vector

All linear combinations of x and y can fill the whole x - y plane.

3 3 Equations, 3 Unknowns

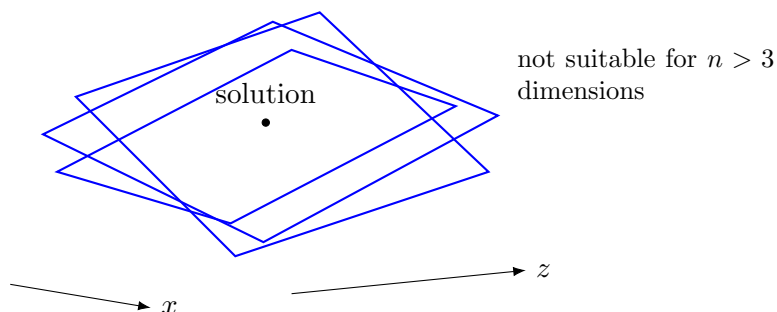
$$\begin{aligned} 2x - y &= 0 \\ -x + 2y - z &= -1 \\ -3y + 4z &= 4 \end{aligned}$$

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

$AX = B$ — matrix form — will be solved in 3 dimensions.

3.1 Row Picture

Each row in the 3×3 problem gives a *plane* in 3 dimensions.



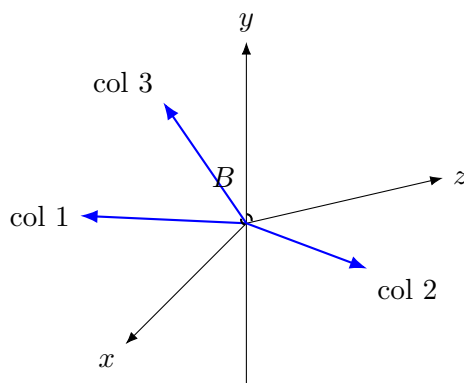
All 3 planes meet at one point, which is the common solution to all 3 equations.

3.2 Column Picture

$$x \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

LHS: linear combination of 3 vectors — each line is a vector in 3 dimensions.

Aim: combine the 3 vectors with correct combination of x , y and z .



$$\left. \begin{array}{l} x = 0 \\ y = 0 \\ z = 1 \end{array} \right\} \text{obvious solution}$$

Large-scale production algorithms use this method to find all possible values of variables with all possible values in the RHS.

Row picture only gave solution to one possible RHS. Column picture can be used to combine any values of variables to produce any possible numeric result.

4 Can We Solve $Ax = b$ for Every b ?

Q: Can we solve $Ax = b$ for every b ?

Q: Do the linear combinations of the columns fill the entire 3-D space?

Yes, yes, for non-singular ($|A| \neq 0$) and invertible matrices.

* What could go wrong while combining 3 vectors and producing a resultant in 3 dimensions?

It fails when all 3 of them lie in the same plane. Solutions/RHS are only restricted to that plane, hence the entire 3-D space can't be filled.

$\implies$ singular and non-invertible case.

Think about 9 equations, 9 unknowns — we will have to do linear combinations of 9 vectors in 9-dimensional space $\implies$ 9 components, 9 rows/columns.

It will yield good results *if* any $[a_R] \neq [a_{R+1}]$, $0 < R < 9$, $R \in \mathbb{I}$ — [] column — otherwise it will fail.

Here we will be restricted [to] several 8-dimensional planes in a 9-D space.

5 Multiplying a Matrix by a Vector

Method 1: Column Method

$Ax = b$ — column method

$$\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}$$

Method 2: Dot Product

Ax — a linear combination of columns of A .